

Web Programming Project Report: Bayern Munich Themed Webpage

1

Introduction

In an era where digital engagement shapes how fans interact with sports, we set out to design a website that not only entertains but educates—an immersive experience centered around the football club Bayern Munich. Both of us, as Munich natives and football enthusiasts, felt compelled to create a space that celebrates the club's heritage while inviting users to engage through interactive games and trivia. Our goal was to blend aesthetics, functionality, and fun, a combination that turned out to be quite challenging but fun.

The project consists of:

1. A homepage to navigate between all the games
2. 5 Mini Games:
 - (a) **4-in-a-row:** A competitive two-player game with customizable match lengths.
 - (b) **Guess the Price (Higher/Lower):** An interactive challenge where players guess whether a footballer's market value is higher or lower.
 - (c) **Memory Game:** A test of memory featuring images of Bayern Munich players.
 - (d) **Cookie Clicker:** A dynamic game with an upgrade system, simulating incremental progress.
 - (e) **Quiz:** A knowledge-based game offering facts about Bayern Munich's history, players, and achievements.
3. A submission form where users have the ability to suggest new games, which notifies us with every submission.

2

Design Philosophy and Aesthetic Choices

From the outset, we aimed to establish a visual identity that reflects Bayern Munich's legacy. We adopted a color scheme rooted in the club's traditional palette—dominantly

red, complemented by white, black and gold accents—echoing the team’s kit and logo. The website features a loading screen displaying the Bayern crest, which not only reinforces brand consistency but also builds anticipation, creating a seamless transition into each page.

Visual design was not an afterthought but an integral part of the user experience. We carefully selected high-resolution images for player profiles and game interfaces, ensuring that each visual element contributes to immersion rather than detracts from it. This attention to detail extended to typography, layout choices, and responsive design elements, all of which were tailored to create a cohesive, modern interface.

The decision to forgo certain features, such as video integrations, was a deliberate trade-off. We initially explored embedding videos—such as celebratory clips at the end of games—but faced persistent issues: YouTube videos often came with unskippable ads, and free downloadable clips lacked the visual quality we desired. Rather than compromise the user experience, we opted to exclude videos, recognizing that restraint can often lead to a more polished and focused final product.

3

Technical Challenges and Problem-Solving

Building five distinct games presented a diverse range of technical challenges, pushing us to refine our problem-solving skills across both front-end and game logic domains.

3.0.1

Game State Management and Variable Persistence

A recurring challenge was managing game state—particularly tracking and persisting scores across sessions. In the 4-in-a-row and Cookie Clicker games, we initially struggled to maintain variables that stored player progress and dynamically updated the interface. After consulting resources like *Stack* (<https://stackoverflow.com/questions/23849365/adding-scores-to-a-variableStack>) *Overflow*, we implemented a system where scores were stored as JavaScript variables, updated through event listeners and conditionals.

For the 4-in-a-row game, we introduced a match threshold feature: players could choose to play until 3 or 5 victories, and the logic adapted accordingly. Implementing this required careful control of the game flow, including resetting the board, updating the score, and determining when a player had won the series.

3.0.2

Scaling Complexity: Cookie Clicker Upgrades

The Cookie Clicker game introduced another layer of complexity. We implemented an upgrade system where players could purchase an *autoclicker* that incrementally increased their score every 1000 milliseconds using the `setInterval()` function. The autoclicker’s strength was dynamically determined by the player’s purchase history, requiring us to design a system that tracked multiple upgrade levels and recalculated their impact on the

game state. The logic behind this was informed by solutions we discovered at <https://onecompiler.com/javascript/3xgrnx46j>.

This challenge underscored a broader theme: game design often demands balancing user agency (choices and upgrades) with system feedback (automatic score increments), requiring careful data management and precise timing.

3.0.3

Sourcing High-Quality Assets

For the Higher/Lower game, we realized that visual quality significantly influenced user engagement. While it might seem like a minor detail, the clarity of player images shaped the entire interaction. We encountered difficulties sourcing high-resolution, license-free images of Bayern players, which required considerable time spent looking for resources. This experience reinforced the importance of visuals in shaping user immersion.

3.0.4

Slideshow Complexity and Manual Navigation

One of the toughest parts of the homepage was building the slideshow. We needed not only smooth automatic transitions (every five seconds via `setInterval()`), but also the ability for users to manually navigate between slides using the previous and next arrows while hovering over the slideshow. To achieve this, we attached `mouseenter` and `mouseleave` event listeners on the `.slideshow-container` element to *clear* the auto-play timer when the mouse is inside, thus pausing the automated cycling, and to *restart* the interval on mouse leave. This gave users full control with the arrows (`.prev-btn` and `.next-btn`), while ensuring the slideshow resumes automatically once they move the cursor away.

3.0.5

Advanced CSS: The Stadium Reveal Effect

One of the most intricate aspects of our design was the homepage “Show Stadium” button. Clicking it reveals a floating image of the Allianz Arena, creating a dynamic visual that anchors the page’s theme. Achieving this required advanced CSS techniques, including pseudo-elements, z-index manipulation, and smooth transitions. Despite extensive research, it was a YouTube tutorial (<https://www.youtube.com/watch?v=YTbCzhDpBHc>) that ultimately helped us implement the desired effect. This process exemplified the iterative nature of problem-solving in web design: trial, error, and leveraging external resources.

4

Reflections: Beyond the Code

This project was as much an exercise in technical execution as it was in teamwork, creativity, and compromise. Working as a duo brought challenges—aligning schedules, dividing tasks, and merging divergent ideas into one project. We used GitHub in order to facilitate

our collaboration, allowing us to track progress, and resolve issues. This really helped us with productivity and quality of our codes.

We also understood that being perfectionism is more a flaw than a quality since sometimes changing a little detail in the website required lots of analysis of the codes and can be time demanding and energy consuming.

Finally, we embraced AI tools as part of our workflow, particularly for debugging and problem-solving. This experience broadened our perspective on modern development practices: knowing how to ask the right questions, interpret resources critically, and integrate external knowledge is as crucial as writing the code itself.

5

Conclusion

Our Bayern Munich-themed website reflects a balanced approach to design, functionality, and consistent theming. We aimed to create a user-friendly platform where visitors can engage with content, learn more about the club, and enjoy an interactive experience.

This project encouraged us to think beyond just writing code. We considered the message we wanted to share, the tone we aimed to set, and the users we were designing for. Overall, building the site was a valuable learning experience that helped us improve our collaboration and technical skills.